

# DEENADAYALAN K

## Software Engineer – AI & Data Platforms

Chennai, India | **Open to relocate: Bangalore / Hyderabad** | +91-8838109535 | deena011197career@gmail.com

Portfolio: portfolio-sage-kappa-46.vercel.app

LinkedIn: linkedin.com/in/deenadayalan-kalaivanan-44679729b | GitHub: github.com/Deenadayalan369

**Serving notice – available to join from 29 Dec 2026**

## PROFESSIONAL SUMMARY

Software Engineer with 3+ years building an enterprise ETL and data validation platform with Python, Node.js, FastAPI, Pandas and RabbitMQ. Benchmarked it reconciling **50 million rows with zero differences** on a single machine, shipped an **LLM feature (OpenAI API)** that turns plain-English business rules into Pandas code, and used Claude Code to build an **MCP server** that lets users operate the platform through a chat interface. Promoted to Software Engineer in 2026; now building agentic, chat-based LLM tools.

## TECHNICAL SKILLS

**Languages:** Python, JavaScript, TypeScript, SQL

**AI / LLM:** OpenAI API, Claude, MCP (Model Context Protocol), LLM Integration, Prompt Engineering, Natural Language to Code

**Backend & APIs:** Node.js, FastAPI, FeathersJS, Express.js, REST APIs, Microservices

**Data Engineering:** ETL Pipelines, Data Reconciliation, Data Validation, Data Quality, Pandas, Dask

**Databases:** PostgreSQL, MySQL, Oracle, MongoDB, Apache Hive

**Frontend:** Angular, React

**Tools:** Git, RabbitMQ, Node-RED, WebSockets, Claude Code

**Evaluated (R&D):** Apache Spark, Trino, Apache Ignite, gRPC, Next.js, Go, Rust, React Flow, shadcn/ui

## PROFESSIONAL EXPERIENCE

**Tristha Global Pvt. Ltd. – Chennai, India**

Sep 2023 – Present

*Software Engineer (Apr 2026 – Present) · Associate Software Engineer (Sep 2023 – Apr 2026)*

Travis – enterprise low-code ETL and data testing platform

- Built a data reconciliation engine (hash-partitioned by join key across RabbitMQ queues and Python workers) that scaled from failing under 1 GB to comparing **two 75 GB files in 7 hours** on one 16 GB machine (proof of concept).
- Benchmarked the platform at **50 million rows**: loaded CSV to MongoDB at **~52,800 rows/sec** and reconciled every row across CSV, DAT, MongoDB and Hive with **zero differences**.
- Shipped an **LLM feature (OpenAI API)** that converts plain-English business rules into Pandas expressions, so non-developers can build transformations without code.
- Used **Claude Code** to build an **MCP (Model Context Protocol) server** for Travis ETL, connected to the MCP client chat window in the new Terra-Sentinel interface, so users can run platform operations through natural language.
- Fixed a critical failure where RabbitMQ dropped validation workers under load, traced through broker logs; a 50M-row DAT-to-Hive validation went from stalling at **17–74%** to **100% complete**.
- Built a Python utility (Pandas, Dask) for a specific client requirement that reconciles DAT and CSV files cell by cell, handling files up to **2 GB**.
- Migrated the core transformation engine from Node.js/Formula.js to a Python (Pandas, FastAPI) microservice.
- Migrating the legacy UI to the new Terra-Sentinel Angular 20 interface (standalone components, signals), using **Claude Code** for AI-assisted development and refactoring.
- Designed a message-driven architecture: custom Node-RED nodes wired through RabbitMQ to Python workers, running pipelines independently of the central server and ending bulk-load memory crashes.
- Designed a metadata-driven data quality framework with configurable duplicate, null and format checks.
- Built a General Ledger (GL) validation engine that evaluates user-defined business-rule expressions and generates Pass/Fail results automatically.
- Led R&D of the Joiner node for large-dataset joins; streamed live pipeline status to the Angular dashboard over WebSockets.

## Freelance Software Developer – Self-Employed

Sep 2022 – Aug 2023

- Built a Hospital and Patient Management System from scratch (MongoDB, Express.js, React, Node.js): REST APIs, authentication, database schema and role-based access control for patients and clinical staff.
- Migrated an enterprise Angular application (Blockyt) from Angular 8 to Angular 14, rewriting legacy code to support newer npm packages and modernise the UI.

## RESEARCH & DEVELOPMENT (TRISTHA GLOBAL)

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- Evaluated Apache Ignite and Trino as federated query engines for heterogeneous joins, testing cross-source joins on 1 GB of Oracle and MongoDB data.
- Compared Dask, Apache Spark and RabbitMQ-based worker queues for scaling ETL processing beyond a single machine; RabbitMQ worker queues were adopted in the platform.
- Built a **gRPC** dataflow prototype as an alternative to REST APIs and WebSockets for moving data between services; after comparing it with RabbitMQ, RabbitMQ was selected for the platform.
- Evaluated React Flow and shadcn/ui for a next-generation visual pipeline builder.
- Analysed Next.js, Go and Rust backend structures for a next-generation platform architecture, including a Go API layer and a Rust execution worker.

## PROJECTS

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### DataNerva – Prompt-Based Data Validation Engine

In progress

- Building a data validation engine driven entirely by natural-language prompts: users describe checks in plain English and the engine runs them, with no SQL queries or code.

### AI Document Manager

In progress

- Building a document storage and viewing app where users upload files and get insights by asking questions in plain language.

### AI Communication Assistant

In progress

- Building a personal assistant that connects communication channels and returns summaries on request through plain-language prompts, to improve daily productivity.

### CuraSynk – Offline-First Family Health Records App

Flutter, SQLite, Android

GitHub: [github.com/Deenadayalan369/CuraSynk](https://github.com/Deenadayalan369/CuraSynk)

- Built and distributed an offline-first Android app (v1.0.1; signed APK, Play Store closed testing) with family profiles, medical history tracking, medication reminders with local notifications, and an offline PDF/image document viewer.

## EARLIER EXPERIENCE

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### Engineer – Condition Monitoring, MVS Acmei Technology Pvt. Ltd.

Aug 2021 – Aug 2022

- Collected and analysed vibration data from industrial rotating equipment to detect faults, and wrote diagnostic reports with predictive-maintenance recommendations to reduce downtime.

### Graduate Engineering Trainee, Sri Sai Equipments Pvt. Ltd.

Dec 2020 – Jul 2021

- Supported operations and maintenance teams with inspection, troubleshooting and installation of industrial machinery.

### Field Service Technician, Self-Employed

Aug 2019 – Jul 2020

- Installed and configured billing software, POS systems, kiosks and Windows systems at retail and hotel client sites.

## EDUCATION

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### B.E. Mechanical Engineering – Saranathan College of Engineering (Anna University)

2015 – 2019

Tamil Nadu, India | CGPA 7.62/10

Languages: English, Tamil, Hindi